دستورات لازم برای حل سوالات و ترسیم نمودارها

تبدیل DFT سیگنال xn با تعداد N نمونه:

```
function [Xk] = dft(xn,N)
% Computes Discrete Fourier Transform
% ----------------------------------
% [Xk] = dft(xn,N)
% Xk = DFT coeff. array over 0 <= k <= N-1
% xn = N-point finite-duration sequence
%  N = Length of DFT
%
n = [0:1:N-1];              % row vector for n
k = [0:1:N-1];              % row vecor for k
WN = exp(-j*2*pi/N);        % Wn factor
nk = n'*k;                  % creates a N by N matrix of nk values
WNnk = WN .^ nk;            % DFT matrix
Xk = xn * WNnk;             % row vector for DFT coefficients

function [xn] = idft(Xk,N)
% Computes Inverse Discrete Transform
% ----------------------------------
% [xn] = idft(Xk,N)
% xn = N-point sequence over 0 <= n <= N-1
% Xk = DFT coeff. array over 0 <= k <= N-1
%  N = length of DFT
%
n = [0:1:N-1];              % row vector for n
k = [0:1:N-1];              % row vecor for k
WN = exp(-j*2*pi/N);        % Wn factor
nk = n'*k;                  % creates a N by N matrix of nk values
WNnk = WN .^ (-nk);         % IDFT matrix
xn = (Xk * WNnk)/N;         % row vector for IDFT values
```

مثال:

Compute the 4-point DFT of $x(n)$.

$$x(n) = \begin{cases} 1, & 0 \le n \le 3 \\ 0, & \text{otherwise} \end{cases}$$

```
>> x = [1,1,1,1]; N = 4;  X = dft(x,N);
>> magX = abs(X), phaX = angle(X)*180/pi
magX =
    4.0000    0.0000    0.0000    0.0000
phaX =
         0 -134.9810  -90.0000  -44.9979
```

بجای دستورات بالا می‌توانید از دستور $fft(x,N)$ و $ifft(x,N)$ نیز استفاده کنید.

تبدیل DFT سیگنال های پیوسته:

$$x(n) = \cos(0.48\pi n) + \cos(0.52\pi n)$$

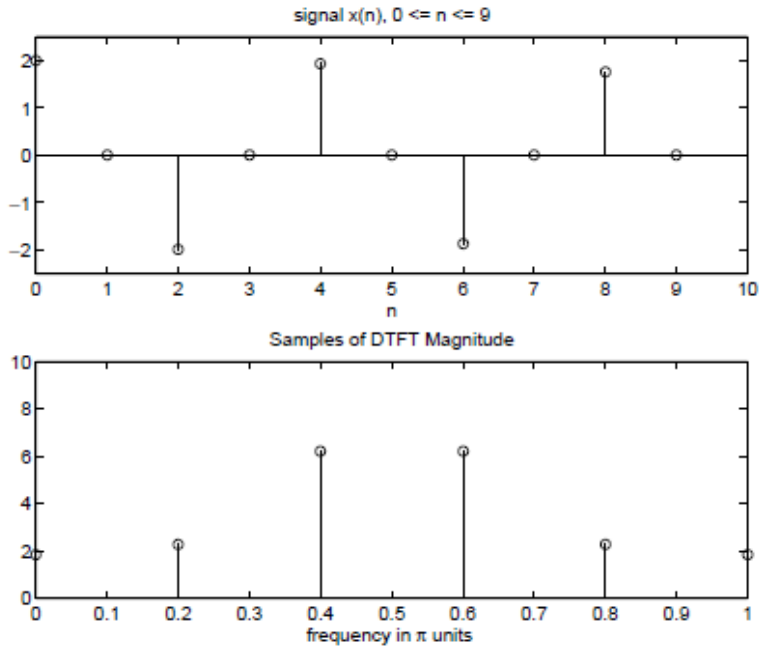
We want to determine its spectrum based on the finite number of samples.

a. Determine and plot the discrete-time Fourier transform of $x(n)$, $0 \leq n \leq 10$.

a. We can first determine the 10-point DFT of $x(n)$ to obtain an estimate of its discrete-time Fourier transform.
MATLAB Script:

```
>> n = [0:1:99]; x = cos(0.48*pi*n)+cos(0.52*pi*n);
>> n1 = [0:1:9] ;y1 = x(1:1:10);
>> subplot(2,1,1) ;stem(n1,y1); title('signal x(n), 0 <= n <= 9');xlabel('n')
```

```
>> Y1 = dft(y1,10); magY1 = abs(Y1(1:1:6));
>> k1 = 0:1:5 ;w1 = 2*pi/10*k1;
>> subplot(2,1,2);stem(w1/pi,magY1);title('Samples of DTFT Magnitude');
>> xlabel('frequency in pi units')
```



خاصیت شیفت دایروی:

```
function y = cirshftt(x,m,N)
% Circular shift of m samples wrt size N in sequence x: (time domain)
% ------------------------------------------------------------------
% [y] = cirshftt(x,m,N)
% y = output sequence containing the circular shift
% x = input sequence of length <= N
% m = sample shift
% N = size of circular buffer
%  Method: y(n) = x((n-m) mod N)
% Check for length of x
if length(x) > N
      error('N must be >= the length of x')
end
x = [x zeros(1,N-length(x))];
n = [0:1:N-1];  n = mod(n-m,N);  y = x(n+1);
```

مثال:

Given an 11-point sequence $x(n) = 10\,(0.8)^n\,, \quad 0 \le n \le 10$, determine and plot $x\,((n-6))_{15}$.

```
>> n = 0:10; x = 10*(0.8) .^ n;  y = cirshftt(x,6,15);
>> n = 0:14; x = [x, zeros(1,4)];
>> subplot(2,1,1); stem(n,x); title('Original sequence')
>> xlabel('n'); ylabel('x(n)');
>> subplot(2,1,2); stem(n,y);
>> title('Circularly shifted sequence, N=15')
>> xlabel('n'); ylabel('x((n-6) mod 15)');
```

